



Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no.....

Name of the Course: B. Tech.

Semester: II

Name of the Paper: Engineering Chemistry

Paper Code: TCH-201

Time: 1.5 hour

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub-questions
- (ii) Each question carries 10 marks.

Q1. (10 Marks)

- a. Draw the MOT diagram of O_2 molecule. Arrange O_2 , O_2^+ , O_2^- and O_2^{2-} in increasing order of stability. (CO1)

OR

- b. Describe band theory of metallic bond with the help of suitable example. (CO1)

Q2. (10 Marks)

- a. Define hydrogen bonding and its conditions. With the help of hydrogen bonding explain why o-nitrophenol can be steam distilled while p-nitrophenol cannot (CO1)

OR

- b. Explain the following:

- i. Principle of spectroscopy and its advantages
- ii. With the help of MOT diagram show the number of non-bonding molecular orbitals in valence shell of HF molecule. (CO1)

Q3. (10 Marks)

- a. The hardness of 40,000 liters of sample water was removed by passing it through a zeolite softener. The softener then required 150 litres of NaCl solution containing 100g/l of NaCl. Calculate the hardness of sample water. (CO2)

OR

- b. Describe the requirement of lime and soda for removing calcium and magnesium salts hardness. Also give the reactions. What are the disadvantages of lime soda process? (CO2)

Q4. (10 Marks)

- a. Explain the Ion-Exchange method of water treatment with the help of diagram. Also discuss the regeneration process and advantages of this method. (CO2)

OR

- b. Explain, why hardness of water is expressed in terms of $CaCO_3$ equivalents. A water sample on analysis was found to consist the following impurities- $Mg(HCO_3)_2 = 32.4$ ppm; $Ca(HCO_3)_2 = 14.6$ ppm; $MgSO_4 = 6.8$ ppm; $CaCl_2 = 9.5$ ppm; $NaCl = 23.4$ ppm. Calculate the temporary and permanent hardness of water. (CO2)

Q5. (10 Marks)

- a. Write a short note on the following: i. Scales and Sludges. ii. Caustic embrittlement (CO2)

OR

- b. Explain the following:

- i. Difference between BMO and ABMO
- ii. Nanomaterials with their applications (CO1)